Expository Writing Assignment

Name: Mahnoor Fatima

Roll No: 1048

Class: 3M CS

Submitted to: Ms Shahida Parveen

Exercise 2: Writing a Diagnostic Paragraph

Field of Study: Computer Science

Chosen Concept: Algorithms

Real-World Application: GPS Navigation Systems

Diagnostic Paragraph:

Algorithms are one of the most important concepts in Computer Science, and they have a powerful real-world application in GPS navigation systems like Google Maps and Waze. Every time a person opens a map app and searches for the shortest route to a destination, a shortest-path algorithm is quietly working in the background to make the journey possible. For example, Dijkstra's Algorithm, which was developed by the computer scientist Edsger Dijkstra in 1956, is widely used by navigation apps to find the fastest route between two points by treating the road network as a graph of nodes and edges. Modern GPS systems also use the A* (A-star) algorithm, which improves on Dijkstra's method by using smart heuristics to find the best routes more quickly, especially over long distances. These algorithms process real-time traffic, road closures, and alternate paths within seconds, which is why a driver stuck in traffic can instantly see a faster route on the screen. The same shortest-path idea is also used in airline flight planning, food delivery apps like Foodpanda, and even in how data packets travel across the internet. In conclusion, algorithms are far more than classroom theories; they are the invisible engines that guide us to our destinations and show how computer science quietly shapes modern life.